

● EASY

● ALGEBRA

● ANSWER KEY

Linear inequalities in one or two variables

02

10 answers · Rationale-rich · Teach from doc

Q1**D**

D. $y \geq 36$

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in the form $y = b$, where $(0, b)$ is the y -intercept of the line in the xy -plane. For the graph shown in the xy -plane, the boundary line passes through the points $(-15, 36)$ and $(0, 36)$. Thus, the equation of the boundary line is $y = 36$. Since the shaded region represents all the points on and above this boundary line, it follows that the graph represents all the solutions to the inequality $y \geq 36$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q2**A**

A. $f + h \leq 14$

Choice A is correct. It's given that the cleaning service cleans both offices and homes, where f is the number of offices and h is the number of homes the cleaning service can clean per day. Therefore, the expression $f + h$ represents the number of places the cleaning service can clean per day. It's also given that the cleaning service can clean at most 14 places per day. Since $f + h$ represents the number of places the cleaning service can clean per day and the service can clean at most 14 places per day, it follows that the inequality $f + h \leq 14$ represents this situation.

Choice B is incorrect. This inequality represents a cleaning service that cleans at least 14 places per day.

Choice C is incorrect. This inequality represents a cleaning service that cleans at most 14 more offices than homes per day.

Choice D is incorrect. This inequality represents a cleaning service that cleans at least 14 more offices than homes per day.

Q3**D**

D. $150 \leq s \leq 170$

Choice D is correct. It's given that during a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. It's also given that s represents the cruising speed, in miles per hour, during this portion of the flight. It follows that the airplane's cruising speed, in miles per hour, was at least 150, which means $s \geq 150$, and was at most 170, which means $s \leq 170$. Therefore, the inequality that best represents this situation is $150 \leq s \leq 170$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4**A**

A. $15s + 25p \leq 120$

Choice A is correct. Since the cost of each shirt is \$15 and Geoff buys s shirts, the expression $15s$ represents the amount Geoff spends on shirts. Since the cost of each pair of pants is \$25 and Geoff buys p pairs of pants, the expression $25p$ represents the amount Geoff spends on pants. Therefore, the sum $15s + 25p$ represents the total amount Geoff spends at the store. Since Geoff can spend at most \$120 at the store, the total amount he spends must be less than or equal to 120. Thus, $15s + 25p \leq 120$.

Choice B is incorrect. This represents the situation in which Geoff spends at least, rather than at most, \$120 at the store. Choice C is incorrect and may result from reversing the cost of a shirt and that of a pair of pants. Choice D is incorrect and may result from both reversing the cost of a shirt and that of a pair of pants and from representing a situation in which Geoff spends at least, rather than at most, \$120 at the store.

Q5**B**

B. $1,510 \leq d \leq 4,130$

Choice B is correct. It's given that during spring migration, a dragonfly traveled a minimum of 1,510 miles and a maximum of 4,130 miles between stopover locations. It's also given that d represents a possible distance, in miles, this dragonfly traveled between stopover locations. It follows that the inequality $1,510 \leq d \leq 4,130$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which a dragonfly traveled a maximum of 1,510 miles between stopover locations.

Choice C is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of 4,130 miles between stopover locations.

Choice D is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of 4,310 miles and a maximum of 5,640 miles between stopover locations.

Q6**C**

C. I and II only

Choice C is correct. Substituting $(1,1)$ into the inequality gives $5(1) - 3(1) < 4$, or $2 < 4$, which is a true statement. Substituting $(2,5)$ into the inequality gives $5(2) - 3(5) < 4$, or $-5 < 4$, which is a true statement. Substituting $(3,2)$ into the inequality gives $5(3) - 3(2) < 4$, or $9 < 4$, which is not a true statement. Therefore, $(1,1)$ and $(2,5)$ are the only ordered pairs shown that satisfy the given inequality.

Choice A is incorrect because the ordered pair $(2,5)$ also satisfies the inequality. Choice B is incorrect because the ordered pair $(1,1)$ also satisfies the inequality. Choice D is incorrect because the ordered pair $(3,2)$ does not satisfy the inequality.

Q7**C****C. 97.9°F**

Choice C is correct. Normal body temperature must be greater than or equal to 97.8°F but less than or equal to 99°F . Of the given choices, 97.9°F is the only temperature that fits these restrictions.

Choices A and B are incorrect. These temperatures are less than 97.8°F , so they don't fit the given restrictions. Choice D is incorrect. This temperature is greater than 99°F , so it doesn't fit the given restrictions.

Q8**B****B. 3.7**

Choice B is correct. It's given that monarch butterflies can fly only with a body temperature of at least 55.0 degrees Fahrenheit $^{\circ}\text{F}$. Let x represent the minimum increase needed in the monarch butterfly's body temperature to fly. If the monarch butterfly's body temperature is 51.3°F , the inequality $51.3 + x \geq 55.0$ represents this situation. Subtracting 51.3 from both sides of this inequality yields $x \geq 3.7$. Therefore, if the monarch butterfly's body temperature is 51.3°F , the minimum increase needed in its body temperature, in $^{\circ}\text{F}$, so that it can fly is 3.7.

Choice A is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is 53.7°F , not 51.3°F .

Choice C is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is 50.0°F , not 51.3°F .

Choice D is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is 48.7°F , not 51.3°F .

Q9**4**

The correct answer is 4. The equation $60h + 10 \leq 280$, where h is the number of hours the boat has been rented, can be written to represent the situation. Subtracting 10 from both sides and then dividing by 60 yields $h \leq 4.5$. Since the boat can be rented only for whole numbers of hours, the maximum number of hours for which Maria can rent the boat is 4.

Q10**D**

D. $25 + 10t \leq 75$

Choice D is correct. The cost of the rental fee depends on the number of hours the surfboard is rented. Multiplying t hours by 10 dollars per hour yields a rental fee of $10t$ dollars. The total cost of the rental consists of the rental fee plus the 25 dollar service fee, which yields a total cost of $25 + 10t$ dollars. Since the person intends to spend a maximum of 75 dollars to rent the surfboard, the total cost must be at most 75 dollars. Therefore, the inequality $25 + 10t \leq 75$ represents this situation.

Choice A is incorrect. This represents a situation where the rental fee, not the total cost, is at most 75 dollars.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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